

above 5 GHz.

Working of a microprocessor is explained here with the help of the 8086 microprocessor architecture. The 8086 is a 16-bit processor. The figure shows its various units. The 8086 CPU comprises two functional units working in parallel—bus interface unit (BIU) and execution unit (EU)—thereby increasing the processing speed of the CPU.

BIU. The BIU fetches the instructions and data bytes from the memory and passes them on to the EU for further decoding and executing. It also reads data from the port and writes data to the memory and the port.

Sub-units of BIU are the instruction queue and segment registers. Data bytes are fetched in advance and stored in a first-in-first-out register called instruction queue. The address of the memory bytes that need to be accessed is generated with the help of segment registers. One of the segment registers called the code segment (CS) is used to store the segment address of instructions.

EU. It tells the BIU which memory location to access and what operation to perform. It involves decoding the information contained in the bits of the instruction word. EU has following sections:

Instruction decoder. Instructions fetched from the instruction queue are decoded and the action to be performed is determined.

Arithmetic logic unit (ALU). It performs the arithmetic and logical operation.

Control circuitry. It controls all actions so that different circuits perform the allotted task in the same time-frame.

Registers. These include special-purpose as well as general-purpose registers.

BIU and EU are connected with the help of buses. There are three types of buses:

Address bus, which contains the address of the memory location that needs to be accessed.

Data bus, which has the data to be fetched/written from/to the memory location addressed.

Control bus, which governs the information transfer through control

signals, indicating the type of operation to be performed on the memory.

In short, to run the program, the processor follows the following sequence:

1. Generate the address and fetch the instruction
2. Decode the instruction with the help of op-code contained in the instruction word
3. Fetch the operands required for execution
4. Execute the instruction

Q2. I liked the 'Claytronics' article published in EFY Nov. 2009. It would be extremely helpful to me if you could explain:

1. Basics of claytronics
2. Catoms and their overall functioning
3. How can these be related to 3D assembly of magnetic and semiconducting materials, abiding by all the rules and regulations?

Abhiroop Dutta

Through e-mail

A2. 1. Claytronics is an abstract future concept that combines nanoscale robotics and computer science to create individual nanometre-scale computers called 'claytronic atoms,' or 'catoms,' that can interact with each other to form tangible 3D objects that a user can interact with. This idea is more broadly referred to as programmable matter.

Claytronics has the potential to greatly affect many areas of daily life, including telecommunication, human-computer interfaces and entertainment.

2. Catoms are similar in nature to a nanomachine, but with greater power and complexity. While these are microscopic individually, they bond and work together on a larger scale. Catoms can change their density, energy levels, state of being and other characteristics using thought alone.

3. Through hardware engineering projects, researchers in the Carnegie Mellon-Intel Claytronics Project investigate the effects of scale on micro-electromechanical systems and model concepts for manufacturable, nanoscale modular

robots capable of self-assembly. Catoms created from this research to populate claytronic ensembles will be less than a millimetre in size, and the challenge in designing and manufacturing them draws the CMU-Intel Research team into a scale of engineering where devices have never been built. The team designing these devices is testing concepts that cross the frontiers of computer science, modular robotics and systems nanotechnology.

At the current stage of design, claytronics hardware operates from macroscale designs with devices that are much larger than the tiny modular robots that set the goals of this engineering research. Such devices are designed to test concepts for sub-millimetre-scale modules and to elucidate crucial effects of the physical and electrical forces that affect nanoscale robotics. With the millimetre-scale modular robot, claytronics will demonstrate the feasibility of manufacturing catoms in the quantities needed to produce dynamic 3-dimensional representations of original objects.

The subject itself is still under development and a lot of practical work is going on at various research institutes and universities world over to realise it practically from micro to macro level. For further details, readers can visit the following websites:

- www.cmu.edu/corporate/news/2007/features/claytronics.shtml
- www.cs.cmu.edu/~claytronics/
- <http://en.wikipedia.org/wiki/Claytronics>
- <http://future.wiki.com/wiki/Claytronics>
- <http://replicatorinc.com/blog/2009/07/claytronics-programmable-matter/>
- www.cs.cmu.edu/~claytronics/hardware/index.html

Answer 1 compiled by Col (Retd) N.C. Pande and answer 2 by Dr Suram Singh Verma (the author of 'Claytronics' article)

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